

Credit Risk Modeling

Predicting Mortgage Defaults in MBS Pools

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The Challenge



Market Scale

\$12 Trillion

MBS Market Size



Crisis Impact

2008 Financial Crisis

Triggered by underestimated
defaults



**Goal: Predict loan-level defaults with
accuracy and transparency**

Approach: Logistic Regression



Regulatory Acceptance

Interpretable for compliance



Coefficient Explainability

Clear risk drivers



Production Ready

Easy calibration & deployment

Data: Freddie Mac Loan Performance



20.32M

Records



20

Features
Engineered



0.8-1%

Default Rate



Period: 2013-2024 | ⚠ Challenge: Extreme Class
Imbalance (1:99)

Rigorous Validation Strategy

Training

≤ Dec 2023

7.87M records

Validation

Jan-Jun 2024

295K records

Test

Jul 2024+

12.2M records

✓ **Key Strength: True out-of-time testing on completely unseen future data**

Feature Engineering

Borrower Profile

- Credit Score
- Original LTV
- DTI Ratio
- Number of Borrowers

Loan Characteristics

- Interest Rate
- Loan Age
- Remaining Term Ratio

Risk Flags

- High DTI (>45%)
- High LTV (>95%)
- Low FICO (<660)
- Negative Equity

Macro at Origination

- 30-yr Mortgage Rate
- Unemployment
- HPI YoY Change
- Fed Funds Rate

Modeling Pipeline

1

Preprocessing

Median imputation → Standard scaling → One-hot encoding



2

Model Training

Logistic Regression with L2 regularization (class_weight=None, solver='lbfgs')



3

Hyperparameter Tuning

Cross-validated grid search on training data



4

Calibration

Platt scaling on validation set for reliable probabilities

Out-of-Time Performance (H2 2024+)

70.05%

ROC-AUC

4.66%

Precision
@ 95th %ile

23.8%

Recall
@ 95th %ile

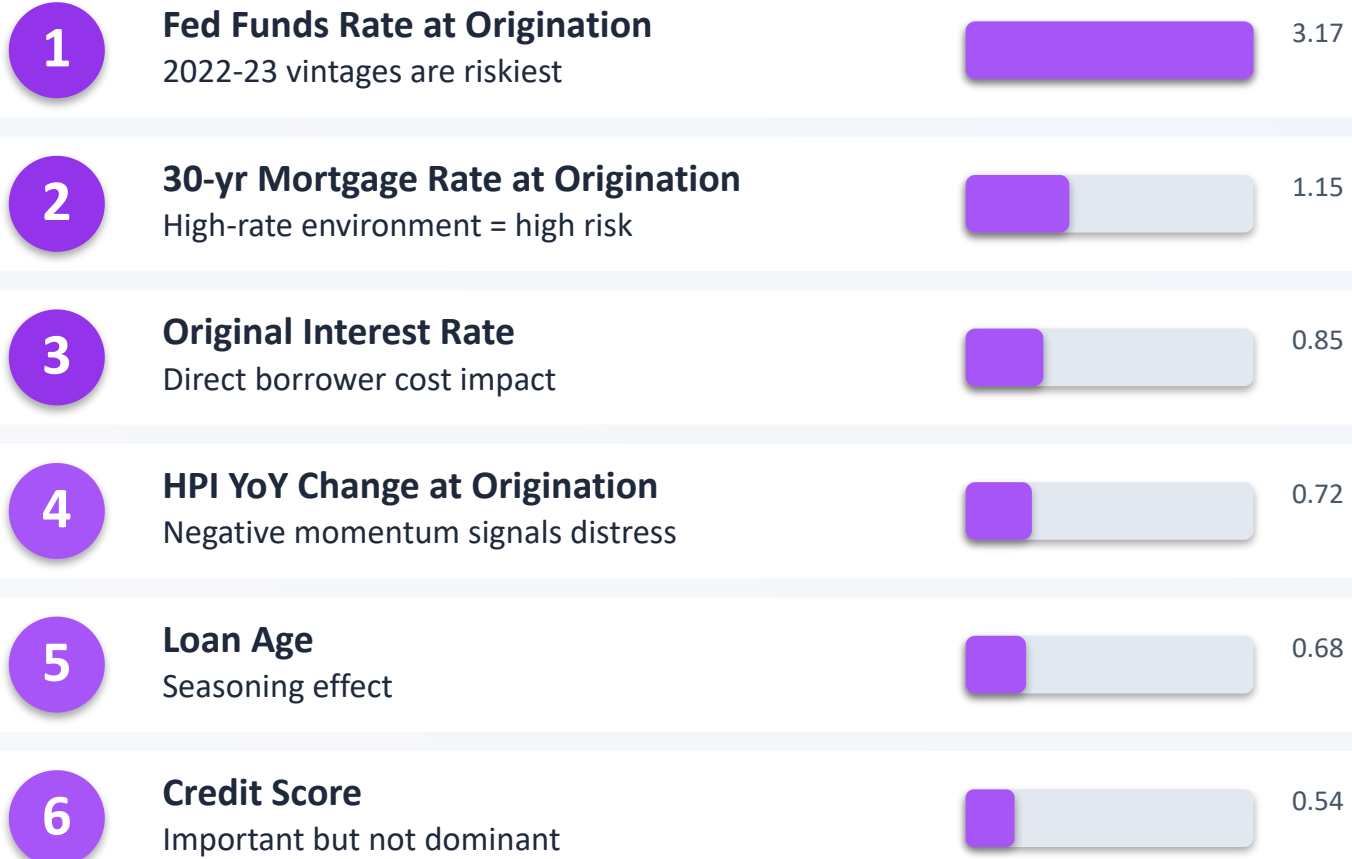
6x

Lift vs
Baseline



Model successfully ranks future defaults in a rising-rate environment

Key Risk Drivers (SHAP Analysis)



Macro conditions at origination matter MORE than borrower credit score

Economic Insights



2022-2023 vintages show significantly higher risk due to rate environment



Origination macro conditions dominate over borrower characteristics



Negative house price momentum is a strong default predictor



Judicial foreclosure states show moderately elevated risk



Credit score matters, but ranks 6th among all drivers

Limitations & Future Enhancements



Current Limitations

- Current macro variables not incorporated (only origination)
- No competing risk modeling with prepayment
- Missing delinquency burnout and payment shock features

✓ Future Enhancements

- Add dynamic unemployment and HPI changes
- Implement LightGBM/XGBoost for higher AUC
- Multi-period survival or cure modeling
- Incorporate prepayment as competing risk

Key Takeaways

- 1 Built transparent default model on 20M+ real loan-month records
- 2 Strict temporal split ensures credible out-of-time performance
- 3 Macro origination conditions (Fed Funds & mortgage rates) dominate risk
- 4 Production-ready: pickled pipeline + 95th percentile threshold
- 5 End-to-end data science: engineering → modeling → validation → insights

Thank You

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Questions?